Bal Krishna Kharal

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Profile

Civil Engineering Ph.D. student specializing in building energy modeling, HVAC system design, renewable energy systems, simulation, and AI integration in simulation workflows. Experienced in HVAC design, load calculations, commissioning, cost estimation, field investigations, and technical report preparation for construction projects. Interested in HVAC design, energy audits, building performance, and construction project management. I learn quickly, enjoy teamwork, and like solving challenging problems through practical, energy-efficient engineering solutions.

Education

Ph.D. in Civil Engineering, North Dakota State University
Fargo, ND | Dec 2027 (Expected)
Master of Science in Construction Management, North Dakota State University
Fargo, ND | May 2026
Bachelor of Engineering in Industrial Engineering, Thapathali Campus, Tribhuvan University
Kathmandu, Nepal | July 2022

Relevant Coursework (HVAC & Construction Field)

Construction Cost Estimating; Advanced Project Planning and Control; Construction Management; Decision Making and Risk Analysis; HVAC; Renewable Energy Technology; Applied Statistics;

Professional Experience

Graduate Research Assistant | North Dakota State University (Jan 2024 – Present)

- **Building digital twin (U.S. Department of Defense project):** Develop and calibrate an EnergyPlus–Modelica building model using field measurements and sensitivity analysis. Integrate AI into simulation tools to automate workflows and support real-time data exchange between the building and model.
- **Renewable energy and heat pumps:** Investigate renewable energy integration and evaluate ground-source and multi-source heat pump systems to improve building energy performance.
- **179D energy modeling:** Conducted energy modeling for two large high schools in Minnesota to support Section 179D evaluations.
- **MnDOT rest-area energy upgrades:** Conducted site visits and HVAC condition assessments, developed simulation models using field data, and performed cost-benefit analyses of energy upgrades.

HVAC Engineer | Thermotech Pvt. Ltd. | Kathmandu, Nepal (Apr 2022–Jan 2024)

- Led HVAC design, bill of quantities (BOQ) preparation, and installation for residential and commercial facilities, including kitchen ventilation, cold storage, and operating room ventilation.
- Performed load calculations, equipment selection, duct and piping design using AutoCAD, and cost-benefit comparisons of system options.
- Worked primarily with ducted HVAC, variable refrigerant flow (VRF), split systems, and energy recovery ventilation (ERV), with additional experience on chiller plant projects.
- Coordinated with architects, contractors, and facility managers to support project delivery within budget and schedule.

HVAC Intern | Thermotech Pvt. Ltd. | Kathmandu, Nepal (Dec 2021–Feb 2022)

- Assisted with HVAC layouts in AutoCAD and Revit, load calculations, equipment selection, ductwork design.

Technical Skills

AutoCAD, AutoCAD MEP, SolidWorks, EnergyPlus, OpenModelica, OpenStudio, TRNSYS, Python, MATLAB, Minitab, statistical analysis, cost-benefit analysis, MS Office, MS Project

Publication and Presentations

- "Development of a Multi-Source Heat Pump System Using EnergyPlus/Modelica Co-Simulation Framework for Residential Buildings." ASHRAE Annual Conference, Phoenix, AZ, 2025.
- "Energy-Saving Potential of Sustainable Energy Building Upgrades at Safety Rest Areas in Minnesota." ASHRAE Winter Conference, 2026.
- "An LLM-Agent-Assisted Calibration Framework Integrating Morris Sensitivity Analysis and Genetic Algorithm Optimization for a Field-Monitored Residential Building EnergyPlus Model". Submitted Journal
- Presented: ASHRAE Annual Conference 2025 (oral). Attended ASHRAE Winter 2025 and SimBuild 2024.

Memberships and Leadership

- ASHRAE Student Member; Sigma Lambda Chi Member; Finance Committee Member and College of Engineering Representative, NDSU Graduate Student Council; Secretary, NSA Chautaree, NDSU; Volunteer, NDSU Career Expo (Spring 2024, Fall 2025) & Homecoming Parade (2024).